

Recall from last class:

- If A is an $n \times n$ matrix, then $\lambda \in \mathbb{R}$ is an **eigenvalue** of A if there exists a non-zero vector v , called an **eigenvector** corresponding to λ , such that

$$Av = \lambda v.$$

- If λ is an eigenvalue of an $n \times n$ matrix A , then the set of all solutions v to $Av = \lambda v$ is the **eigenspace** of A corresponding to λ .
- **Theorem 1:** The eigenvalues of a triangular matrix are the diagonal entries.
- **Theorem 2:** If $v_1, \dots, v_p$ are eigenvectors corresponding to distinct eigenvalues $\lambda_1, \dots, \lambda_p$ of an $n \times n$ matrix A , then $v_1, \dots, v_p$ are linearly independent.
- If A is an $n \times n$ matrix, then the characteristic polynomial of A , denoted by $p_A(\lambda)$, is given by

$$p_A(\lambda) = \det(A - \lambda I_n).$$

- λ is an eigenvalue of an $n \times n$ matrix A if and only if λ is a zero of the characteristic polynomial.
- If A and B are similar matrices, then they have the same characteristic polynomials and hence the same eigenvalues (with the same algebraic multiplicities).
- An $n \times n$ matrix A is **diagonalizable** if A is similar to a diagonal matrix.
- **Diagonalization Theorem:** An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors. Moreover, $A = PDP^{-1}$ with D diagonal if and only if the columns of P are n linearly independent eigenvectors of A . In this case the diagonal entries of D are the corresponding eigenvalues of A .
- **Theorem 6:** An $n \times n$ matrix A with n distinct eigenvalues is diagonalizable.

Theorem 7: Let A be an $n \times n$ matrix with distinct eigenvalues $\lambda_1, \dots, \lambda_p$.

1. For $1 \leq k \leq p$, the dimension of the eigenspace of λ_k is at most the algebraic multiplicity of the eigenvalue λ_k .
2. The matrix A is diagonalizable if and only if the sum of the dimensions of the eigenspaces is n if and only if the characteristic polynomial factors into linear factors and the dimension for each eigenspace is exactly the algebraic multiplicity.
3. If A is diagonalizable and $\mathcal{B}_k$ is a basis for the eigenspace corresponding to λ_k , then $\mathcal{B}_1, \dots, \mathcal{B}_p$ is an eigenbasis for $\mathbb{R}^n$.

1. Prove the theorem.

2. In each part, determine if there is a basis in which the linear transformation is represented by a diagonal matrix.

(a) Let $T : \mathbb{P}_2 \rightarrow \mathbb{P}_2$ given by $T(p) = p(t) + (t + 1)p'(t)$.

(b) Let $T : \mathbb{P}_2 \rightarrow \mathbb{P}_2$ given by $T(p) = p(1) + p'(0)t + (p'(0) + p''(0))t^2$.